

Ferran Alía Castillejos

Data Science & Physical Engineering Undergraduate

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Employment History

- 2022 – 2023  **Robotics Teacher** Taught classes in a “Punt Multimedia” about several technological topics, e.g. Arduino MicroBit and 3D printing & modeling.
- 2022 – . . .  **Volunteer Work** Mentored participants in Computer Science/Robotics workshops, about Linux and 3D printing, among others.
- June 2025 – August 2025  **MLCode Fulltime Internship** Developing an automated framework for evaluating and benchmarking Retrieval Augmented Generation LLM systems.
- 2026 – . . .  **Contributor to TDS** Writing articles for the internet publication “Towards Data Science” [profile]

Education

- 2023 – 2028  **Universitat Politècnica de Catalunya** Data Science and Engineering third-year undergraduate. (8.62/10 average)
-  **Universitat Politècnica de Catalunya** Physical Engineering third-year undergraduate. (8.62/10 average)
-  **CFIS Program**, Recipient of the CFIS (Centre de Formació Interdisciplinària Superior) grant to pursue two careers simultaneously.

Relevant Academic Projects

- Fixed-Wing bird flight sim  Wrote a Python 3D Navier-Stokes solver and compared its results with icoFoam solver by OpenFOAM on custom-modelled geometry.
- Parallel N-Body problem  $O(n^2)$ C++ simulation using OpenMP for parallelism. Comparison with nonparallelized Barnes-Hut $O(n \log(n))$ solution.
- PINN for fluid modeling  Training of a PINN to predict Kármán vortex shedding from synthetic data (high fidelity OpenFOAM simulation).
- Fermi-Dirac distribution  A Montecarlo Markov Chain simulation to numerically derive the FD quantum statistical distribution.
- F1 frontwing optimization  Surrogate learning algorithm to optimize the aerodynamics of an F1 frontwing, on the Marenostrum V supercomputer.
- AI for assisted birth  Model to predict both fetal distress and delivery pathway from cardiotocography, medical context and ultrasound imaging.
- LLM persona research  Methodological comparison of instruction-tuning, fine-tuning, and RL-based approaches for LLM persona adaptation
- Lattice Boltzmann solver  Development and benchmarking of a parallel LBM (D2Q9) solver in C++ with OpenMP

Skills

Languages	■ Native in Spanish and Catalan, professional fluency in English
Programming	■ Python, C++, R, Haskell, MATLAB, among others
Tools	■ OpenFOAM, Linux, CAD, Blender, Git, OpenMP
Misc.	■ Experience teaching, self-study, problem solving, critical thinking, HPC

Awards & Additional Activities

2019 – 2021	■ Algorithms and Programming , summer courses by UPC.
June 2025	■ Physics in Ljubljana , summer school by the University of Ljubljana.
November 2024 & 2025	■ Participant at the Datathon FME
2023 – . . .	■ Cum Laude , on several subjects.
2022 – 2023	■ Graduated With Honors in high school.
2023	■ Mostra de Recerca Jove , my final research project (TR) was chosen to represent the school in the Barcelona “Mostra de Recerca Jove”.